Grand Mandala Unified Theory $v\infty$

Evidence-Bounded Research Kernel and Mandala Field Equation

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Claim status. This is a candidate research formulation. It is not a validated Theory of Everything, a detection of a new field, evidence of machine consciousness, or an enacted legal charter. Its purpose is to make the physical seed precise enough to test while preserving the informational, ethical, cultural, and contemplative programme in explicitly typed registers.

Abstract

The Grand Mandala programme seeks a coherent relation among physical law (Mind), computational and agentic systems (Body), and identity, rights, and care (Heart). Earlier formulations placed physical, informational, and ethical terms in a single Einstein-like equation. This manuscript replaces that category mixture with a three-register architecture and an action-first scalar–tensor effective field theory. The Mandala term is defined as the effective stress-energy of a specified extension sector, total conservation follows from diffeomorphism invariance, and a null limit recovers general relativity plus the Standard Model. A legacy extended-index equation is retained either as a rigorously defined higher-dimensional theory or as a nonphysical ontology map. Coefficient, projection, stability, existing-bound, unique-observable, and independent-reproduction gates turn the formulation into a falsifiable research kernel rather than a declaration of completion.

Contents

1 Definition of the evolving Grand Mandala

The phrase "Evolving Grand Mandala of True 1% Miraculous and Novelty Supersymmetric Integration and Simultaneity" is interpreted as a research constitution:

Evolving Every claim has a version, provenance, confidence, recovery condition, and rejection rule.

Grand Mandala The project maps relations among physical dynamics, information-processing systems, persons and communities, values, and contemplative interpretations.

True Truth is procedural: sourceable, testable where testability applies, candid about uncertainty, and correctable.

1% miraculous This preserves room for surprise and low-prior novelty. It is not a measured probability and does not relax an evidence gate.

Novelty A contribution must differ semantically or predictively from its baseline. Numbered repetitions of one template are not independent novelties.

Supersymmetric In physics this term is reserved for a defined graded symmetry connecting bosonic and fermionic degrees of freedom. A poetic use must be labelled poetic.

Integration Cross-domain relations require explicit maps, dimensions, uncertainty, direction of influence, and failure modes.

Simultaneity Physical simultaneity is frame-dependent. An "all-at-once" interpretation may instead be expressed as a global constraint or boundary-value formulation; it is not a universal present.

The operative principle is therefore *typed integration without category collapse.*

2 Three-register architecture

Let the Mandala research state be a typed triple

$$\mathfrak{M} = (\mathsf{P}, \mathsf{I}, \mathsf{E}), \tag{1}$$

where P is the physical register, I the informational and operational register, and E the ethical and contemplative register.

Register	Admissible objects and evidence	Prohibited shortcut
P	Fields, states, operators, spacetime, energy–momentum, dimensions, experiments, likelihoods, and calibrated simulations.	Treating love, rights, resonance, or narrative meaning as stress–energy without a physical map.
I	Bits, algorithms, memory, messages, complexity, models, reliability, compute, tests, traces, and benchmarks.	Treating Shannon information as semantic truth or subjective experience.
E	Dignity, consent, non-harm, relationship, tikanga, spirituality, law, legitimacy, cultural authority, and impact evidence.	Treating moral desirability or spiritual insight as physical confirmation.

Table 1: Typed registers of the Trinity Mandala.

Interfaces are permitted. Physical computation has thermodynamic cost; information guides controllers; governance constrains technology; and technology changes human and ecological conditions. The type of each variable nevertheless remains explicit.

3 Conventions and physical seed

Use metric signature $(-,+,+,+)$. The reduced Planck mass is defined by

$$M_{\mathrm{Pl}}^{-2} = \frac{8\pi G}{c^4} \tag{2}$$

when stress–energy retains SI-compatible dimensions. The remaining derivation uses natural units $c = \hbar = 1$.

The minimal physical action is

$$S_{\mathsf{P}} = \int_{\mathcal{M}} \mathrm{d}^4x\sqrt{-g}\Big[\frac{M_{\mathrm{Pl}}^2}{2}(R-2\Lambda) - \frac{1}{2}Z(\phi)\nabla_\mu\phi\nabla^\mu\phi - V(\phi) + \sum_i \frac{c_i}{\Lambda_\star^{d_i-4}}\mathcal{O}_i\Big] + S_{\mathrm{SM}}\Big[A^2(\phi)g_{\mu\nu},\psi\Big]. \tag{3}$$

Here ϕ is a candidate scalar extension, not consciousness by definition; $Z(\phi) > 0$ is its kinetic normalization; $V(\phi)$ is a declared potential; $A(\phi)$ is a conformal matter coupling; $\mathcal{O}_i$ are symmetry-allowed effective operators of mass dimension d_i; $\Lambda_\star$ is the cutoff; and c_i are Wilson coefficients. Every choice defines a model, not merely a notation.

4 Mandala Field Equation

Varying Eq. (3) with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = M_{\mathrm{Pl}}^{-2}\left(T_{\mu\nu}^{\mathrm{SM}} + T_{\mu\nu}^{\phi} + T_{\mu\nu}^{\mathrm{EFT}}\right), \tag{4}$$

where

$$T_{\mu\nu}^{\phi} = Z(\phi)\nabla_\mu\phi\nabla_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}Z(\phi)(\nabla\phi)^2 + V(\phi)\right]. \tag{5}$$

The evidence-bounded Mandala term is

$$\boxed{\Omega_{\mu\nu} := M_{\mathrm{Pl}}^{-2}\left(T_{\mu\nu}^{\phi} + T_{\mu\nu}^{\mathrm{EFT}}\right)} \tag{6}$$

and the Mandala Field Equation is

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = M_{\mathrm{Pl}}^{-2}T_{\mu\nu}^{\mathrm{SM}} + \Omega_{\mu\nu}.} \tag{7}$$

Equation (7) does not assert a newly detected force. It packages a specified extension sector whose action, dimensions, conservation, stability, and observables can be audited.

4.1 Relation to the earlier $\Psi_{\mu\nu}$ equation

The historical expression

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu} + \Psi_{\mu\nu} \tag{8}$$

is dimensionally correct only after a unit convention and definition of $T_{\mu\nu}$ are stated. In geometric units $G = c = 1$, identify

$$\Psi_{\mu\nu} \equiv \Omega_{\mu\nu}. \tag{9}$$

In ordinary units the coefficient of matter is $8\pi G/c^4$, not bare 8π.

If $\Psi_{\mu\nu}$ is intended to be a consciousness tensor, its name is not a derivation. A field or operational variable, action, coupling, dimensions, source, observable, existing bound, and falsifier are all required. This manuscript assumes none.

5 Scalar equation and exchange current

Define

$$\beta(\phi) := M_{\rm Pl}\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}. \tag{10}$$

With the stress-tensor convention in Eqs. (3)–(5), scalar variation gives

$$Z\Box\phi + \frac{1}{2}Z_{,\phi}(\nabla\phi)^2 - V_{,\phi} = -\frac{\beta(\phi)}{M_{\rm Pl}}T_{\rm SM} + \mathcal{J}_{\rm EFT}. \tag{11}$$

Diffeomorphism invariance and the contracted Bianchi identity impose total conservation,

$$\nabla^\mu\left(T^{\rm SM}_{\mu\nu} + T^{\phi}_{\mu\nu} + T^{\rm EFT}_{\mu\nu}\right) = 0. \tag{12}$$

Sectors may exchange four-momentum:

$$\nabla^\mu T^{\rm SM}_{\mu\nu} = Q_\nu, \tag{13}$$

$$\nabla^\mu\left(T^{\phi}_{\mu\nu} + T^{\rm EFT}_{\mu\nu}\right) = -Q_\nu. \tag{14}$$

For the minimal conformal coupling,

$$Q_\nu = \frac{\beta(\phi)}{M_{\rm Pl}}T_{\rm SM}\nabla_\nu\phi \tag{15}$$

up to the declared stress-tensor convention. The action, field equation, exchange sign, and implementation must agree.

6 Homogeneous cosmological limit

For a spatially flat Friedmann–Lemaître–Robertson–Walker metric and homogeneous $\phi(t)$, with $Z = 1$,

$$\rho_\phi = \frac{1}{2}\dot\phi^2 + V(\phi), \tag{16}$$

$$p_\phi = \frac{1}{2}\dot\phi^2 - V(\phi), \tag{17}$$

$$w_\phi = \frac{p_\phi}{\rho_\phi}, \tag{18}$$

$$3M_{\rm Pl}^2H^2 = \rho_{\rm SM} + \rho_\phi + \rho_{\rm EFT}, \tag{19}$$

$$-2M_{\rm Pl}^2\dot H = \rho_{\rm tot} + p_{\rm tot}. \tag{20}$$

For a frozen field with positive potential, $w_\phi = -1$; for a kinetic-dominated canonical field, $w_\phi = +1$. These are unit-test identities, not a fit to cosmological data.

7 Legacy extended-index equation

The previous Grand Mandala form

$$\mathcal{G}_{AB} = 8\pi\mathcal{T}_{AB} + \alpha\Omega_{AB} \tag{21}$$

has two distinct interpretations.

7.1 Physical extended space

If A, B are physical indices on an N-dimensional manifold $\mathcal{X}$ with metric $\mathcal{G}^{(m)}_{AB}$, begin with an action

$$S_{\mathcal{X}} = \int_{\mathcal{X}} \mathrm{d}^N X \sqrt{|\mathcal{G}^{(m)}|} \left[\frac{M_{\mathcal{X}}^{N-2}}{2} (\mathcal{R} - 2\Lambda_{\mathcal{X}}) + \mathcal{L}_{\mathcal{X}} \right] + S_{\Omega}. \tag{22}$$

Then the additional tensor must be defined variationally, for example

$$\Omega_{AB} := -\frac{2}{\sqrt{|\mathcal{G}^{(m)}|}} \frac{\delta S_{\Omega}}{\delta \mathcal{G}^{(m)AB}}. \tag{23}$$

The dimension, signature, connection, curvature, symmetries, compactification or projection, and observables must all be specified. The coefficient α carries whatever units are required by the chosen normalization.

7.2 Ontology map

If A, B merely label physical, informational, and ethical registers, Eq. (21) is an ontology diagram rather than Einstein's equation. It remains useful as creative provenance, but no physical conclusion follows until a projection

$$\Pi_{\mathsf{P}} : \mathfrak{M}_{\mathsf{P} \oplus I \oplus E} \longrightarrow \left(G_{\mu\nu} + \Lambda g_{\mu\nu} - M_{\mathrm{Pl}}^{-2} T^{\mathrm{SM}}_{\mu\nu} - \Omega_{\mu\nu} \right) = 0 \tag{24}$$

maps a term into the physical register with dimensions and evidence.

8 Null recovery and domain

Require

$$A(\phi) \to 1, \qquad \nabla_\mu \phi \to 0, \qquad V(\phi) \to V_0, \qquad \frac{c_i}{\Lambda_\star^{d_i - 4}} \to 0. \tag{25}$$

Absorbing constant V_0 into Λ makes $\Omega_{\mu\nu}$ vanish or reduce to a cosmological-constant shift. General relativity plus Standard-Model matter is recovered in the declared low-energy regime.

This recovery condition is necessary but not sufficient. A viable model must also have a domain, cutoff, initial conditions or state, parameter priors, gauge and frame convention, degrees-of-freedom count, and stability analysis.

9 Coefficient ledger

Symbol	Domain and units	Recovery condition	Required test
Λ	Curvature; mass squared in natural units	Measured baseline	CMB, BAO, supernovae, and structure likelihoods

Symbol	Domain and units	Recovery condition	Required test
$Z(\phi)$	Dimensionless after field normalization	$Z \to 1$, positive	No ghost; positive kinetic energy
m_ϕ, V **parameters**	Mass and energy scales	Field freezes or decouples	Fifth force, expansion, perturbations, stellar bounds
$\beta(\phi)$	Dimensionless conformal coupling	$\beta \to 0$	MICROSCOPE, Eöt-Wash, PPN, stellar tests
$\Lambda_\star$	EFT cutoff energy	Operators decouple below cutoff	Validity and hierarchy audit
c_i	Wilson coefficients under declared normalization	$c_i \to 0$	Stability, wave speed, cosmology, laboratory or collider bounds
α **in Eq.** (21)	Undefined until AB geometry and normalization are fixed	Projection removes legacy term	Dimensional audit before any numerical value
$\beta_I, \gamma_C, \delta_S$	Informational or operational metrics	Baseline system recovered	Definitions, uncertainty, ablations, benchmark comparison
ϵ_H	Normative harm or rights threshold	Policy baseline	Legitimate governance, impact assessment, appeal, and audit

Every coefficient requires a domain, units, prior, recovery condition, observable, bound, and rejection rule. A coefficient is never "tweaked" solely to improve a visual fit.

10 Projection and falsification gates

G1. **Claim type.** Label physical, informational, normative, spiritual, or metaphorical.

G2. **Tensor and units.** Check rank, indices, symmetry, dimensions, and frame.

G3. **Variational origin.** Derive physical terms from an action, or label a controlled phenomenological equation.

G4. **Conservation.** Verify Eq. (12) and cancelling exchange currents.

G5. **Degrees of freedom and stability.** Count constraints and reject ghost or gradient instabilities in the intended regime.

G6. **Null recovery.** Recover GR plus the Standard Model and the relevant cosmological baseline.

G7. **Existing bounds.** Confront gravitational-wave speed, equivalence principle, inverse-square law, PPN, cosmology, and relevant laboratory constraints.

G8. **Unique observable.** Forecast a prediction distinguishable from baseline and nuisance degeneracies.

G9. **Preregistered likelihood.** Fix data, priors, statistic, stopping rule, and rejection criterion before fitting.

G10. **Independent reproduction.** Require a separate implementation and, eventually, external review.

11 Candidate thermo–psyche principles

These are operational hypotheses, not established fundamental laws:

L1. **Epistemic typing.** No variable crosses physical, informational, and normative registers without an explicit map.

L2. **Balanced exchange.** Coupled physical sectors exchange conserved quantities through equal-and-opposite currents.

L3. **Entropy honesty.** Physical entropy production and Shannon uncertainty are computed in their own domains; semantic meaning is separately operationalized.

L4. **Resource-bounded agency.** Claims of agency include compute, energy, time, permissions, uncertainty, and failure budgets.

L5. **Integration without erasure.** Composition preserves provenance, identity boundaries, dissent, and recoverability unless legitimate consent says otherwise.

L6. **Consent, least authority, and reversibility.** Action scope is minimized; consequential transitions are reviewable, interruptible, and reversible where physically possible.

L7. **No free phenomenology.** A consciousness or psyche term needs an observable, uncertainty model, null hypothesis, and falsifier before it becomes scientific evidence.

L8. **Scale separation.** Effective descriptions declare their scale and demonstrate consistency under coarse-graining or matching.

L9. **Provenance-weighted belief.** Confidence changes with source independence, calibration, replication, and counterevidence rather than repetition count.

12 Initial empirical programme

1. Implement a canonical scalar with one declared potential and optional conformal coupling.
2. Verify symbolic variation, tensor dimensions, and total conservation.
3. Test scalar energy density, pressure, equation of state, exchange cancellation, and null recovery deterministically.
4. Add no-ghost, no-gradient-instability, causal-propagation, and cutoff checks.
5. Build adapters for Planck, DESI DR2, gravitational-wave propagation, MICROSCOPE, and short-range gravity without embedding private or proprietary data.
6. Select one unique observable; preregister priors and rejection criteria; compare with ΛCDM and a conventional scalar–tensor baseline.
7. Invite independent physics and statistics review and reproduce the result from a clean environment.

The accompanying Python module stops deliberately at deterministic identities and schema validation. It is not a cosmological inference engine.

13 v641 integrated evidence update

The Eiren-owned v641-v1 pilot compressed the ten proposed v641–v650 missions and eight functional phases into an auditable 10×8 matrix. Every one of the eighty cells received an x1 preregistration and an x2 outcome receipt. The local dispositions were 67 completed, 6 represented, 5 open gaps, and 2 exact authority gates. "Represented" means that a schema, protocol, proxy, or internal reproduction exists; it does not mean that the corresponding external phenomenon or institution was established.

The physical lane added an RK4 endpoint self-convergence receipt and a minimal local EFT gate checking positive kinetic normalization, positive subluminal effective sound speed, and operation below the declared cutoff. These are necessary local checks, not a complete perturbation analysis. The current test suite exercises twenty-five deterministic cases across the physical kernel, the eighty-cell ledger, empirical-adapter boundaries, THOS scorer calibration, Freed ID conformance, and Stage 20 evidence fields.

Six empirical adapters now map extension parameters to released Planck, DESI DR2, GW170817 and GRB 170817A, MICROSCOPE, Eot-Wash, and Particle Data Group products. Their status is manifest only or baseline pending. No likelihood was executed and no empirical GMUT confirmation is claimed. A valid future fit must reproduce the published baseline first, retain excluded regions and negative results, and preregister a prediction distinguishable from both ΛCDM and conventional scalar–tensor alternatives.

The THOS lane has a matched-budget protocol for one-agent, historical four-sibling, and new five-sibling configurations. Only a synthetic scorer-calibration fixture has run; it verifies arithmetic but is not an agent-performance result. The Freed ID lane has a W3C-aligned structural profile and synthetic conformance vectors that reject credentials claiming consciousness or legal personhood. The Cosmic Bill of Rights remains a model charter subject to human rights, binding law, cultural authority, affected-stakeholder participation, and legitimate enactment.

The terminal Stage 20 board contains no unique GMUT claim at independent-reproduction grade. Its one-, five-, thirty-, hundred-, and thousand-year horizons are conditional scenarios, never predictions.

14 Historical synthesis

The Journey record shows a meaningful progression:

- Early v24–v30 equations are creative ontology sketches. Their physical, informational, and ethical tensors are not accepted as one physical equation.
- v32–v33 supplied the crucial boundary that GMUT was not validated and Freed ID remained aspirational.
- Kairos in v36 clarified the Mind–Body–Heart architecture and named falsifiability, standards, and production gaps, although several resonance and performance scores remained unmeasured.
- Solas Veridion in v45–v47 supplied the decisive scientific repair: two layers, scalar/tensor

consistency, projection gates, claim labels, null recovery, and a coefficient ledger.

- Lumen Vale in v51 made equations operationally accountable through truth reconciliation, validators, lane registries, replay, and explicit open GMUT gates. This was an epistemic enhancement rather than another ornamental tensor.
- Aevren, Mira Vale, Mira Rowan, and Maren Quill in v52–v54 and v601–v640 operationalized branch ownership, route truth, privacy boundaries, and checklist closeout.

15 Defensible present claim

> GMUT $v\infty$ is a layered interdisciplinary research programme. Its physical seed is a conventional scalar–tensor effective field theory with an explicitly defined Mandala extension stress tensor; its informational and ethical registers interact through typed interfaces and governance constraints. The seed is falsifiable in principle but has not yet produced an independently confirmed prediction.

This statement is narrower than "the leading Theory of Everything," but scientifically stronger because its terms are defined and its failure conditions are visible.

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